$$[\text{Word Indices}] \longrightarrow \mathbf{W_1}\ (\text{Input Embedding}) \longrightarrow (\mathbf{W_{hh}}\ \text{Loop across words} + \text{ReLU}) \longrightarrow \mathbf{W_2}\ (\text{Output Projection}) \longrightarrow \text{Final 3D Vector}$$

$$[\text{Word Indices}] \quad \underbrace{\longrightarrow \mathbf{W_1}}_{\text{FNN / Embedding}} \quad \longrightarrow \underbrace{(\mathbf{W_{hh}}\ \text{Loop across words} + \text{ReLU})}_{\textbf{PURE CORE RECURRENT/LSTM ENGINE}} \quad \underbrace{\longrightarrow \mathbf{W_2}}_{\text{FNN Output}} \quad \longrightarrow \text{Final 3D Vector}$$

Let's run a complete, step-by-step mathematical demonstration using your exact W_1 vocabulary weights to compare two sentences with the **exact same words in a different order**:

- **Sentence 1:** "python writing programming"
- **Sentence 2:** "programming writing python"

To make the LSTM respect word order, we introduce a small **Recurrent Weight Matrix (W_{hh})** of size 2×2 to multiply the previous hidden state, plus ReLU activation (keeping your deep learning style).

Step 0: Defining Our Shared Weights

1. **Word Vectors from your W_1 Table:**
    - python (Index 0): $[0.20, -0.10]$
    - programming (Index 1): $[-0.15, 0.30]$
    - writing (Index 2): $[0.35, -0.15]$
2. **Recurrent Weight Matrix (W_{hh}):** Let's define a clean 2×2 recurrent weight matrix that passes memory forward:

$$W_{hh} = \begin{bmatrix} 0.8 & 0.2 \\ -0.2 & 0.8 \end{bmatrix}$$

3. **Initial Hidden State (h_0):** Start with zero memory: $h_0 = [0.0, 0.0]$.

Sentence 1: "python" $\rightarrow$ "writing" $\rightarrow$ "programming"

Let's process Sentence 1 word-by-word sequentially:

Time Step 1: Read "python"

- Input vector $x_1 = [0.20, -0.10]$
- Calculation: $(W_{hh} \times h_0) + x_1 = [0, 0] + [0.20, -0.10] = [0.20, -0.10]$
- Apply ReLU (turns negatives to 0):

$$h_1 = [0.20, 0.0]$$

Time Step 2: Read "writing"

- Input vector $x_2 = [0.35, -0.15]$
- Multiply previous hidden state (h_1) by W_{hh}:

$$W_{hh} \times h_1 = \begin{bmatrix} 0.8 & 0.2 \\ -0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.20 \\ 0.0 \end{bmatrix} = \begin{bmatrix} 0.16 \\ -0.04 \end{bmatrix}$$

- Add new word vector and apply ReLU:

$$h_2 = \max\left(0, \begin{bmatrix} 0.16 \\ -0.04 \end{bmatrix} + \begin{bmatrix} 0.35 \\ -0.15 \end{bmatrix}\right) = \max\left(0, \begin{bmatrix} 0.51 \\ -0.19 \end{bmatrix}\right) = [0.51, 0.0]$$

Time Step 3: Read "programming" (Final Word)

- Input vector $x_3 = [-0.15, 0.30]$
- Multiply previous hidden state (h_2) by W_{hh}:

$$W_{hh} \times h_2 = \begin{bmatrix} 0.8 & 0.2 \\ -0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.51 \\ 0.0 \end{bmatrix} = \begin{bmatrix} 0.408 \\ -0.102 \end{bmatrix}$$

- Add new word vector and apply ReLU:

$$h_3 = \max\left(0, \begin{bmatrix} 0.408 \\ -0.102 \end{bmatrix} + \begin{bmatrix} -0.15 \\ 0.30 \end{bmatrix}\right) = \max\left(0, \begin{bmatrix} 0.258 \\ 0.198 \end{bmatrix}\right) = [\mathbf{0.258}, \mathbf{0.198}]$$

Final Representation for Sentence 1: $[\mathbf{0.258}, \mathbf{0.198}]$

Sentence 2: "programming" $\rightarrow$ "writing" $\rightarrow$ "python"

Now let's process the exact same words in reverse order:

Time Step 1: Read "programming"

- Input vector $x_1 = [-0.15, 0.30]$
- Calculation with $h_0 = [0, 0]$ and ReLU:

$$h_1 = \max(0, [-0.15, 0.30]) = [0.0, 0.30]$$

Time Step 2: Read "writing"

- Input vector $x_2 = [0.35, -0.15]$
- Multiply previous hidden state (h_1) by W_{hh}:

$$W_{hh} \times h_1 = \begin{bmatrix} 0.8 & 0.2 \\ -0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.0 \\ 0.30 \end{bmatrix} = \begin{bmatrix} 0.06 \\ 0.24 \end{bmatrix}$$

- Add new word vector and apply ReLU:

$$h_2 = \max\left(0, \begin{bmatrix} 0.06 \\ 0.24 \end{bmatrix} + \begin{bmatrix} 0.35 \\ -0.15 \end{bmatrix}\right) = \max\left(0, \begin{bmatrix} 0.41 \\ 0.09 \end{bmatrix}\right) = [0.41, 0.09]$$

Time Step 3: Read "python" (Final Word)

- Input vector $x_3 = [0.20, -0.10]$
- Multiply previous hidden state (h_2) by W_{hh}:

$$W_{hh} \times h_2 = \begin{bmatrix} 0.8 & 0.2 \\ -0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.41 \\ 0.09 \end{bmatrix} = \begin{bmatrix} (0.8 \times 0.41) + (0.2 \times 0.09) \\ (-0.2 \times 0.41) + (0.8 \times 0.09) \end{bmatrix} = \begin{bmatrix} 0.346 \\ -0.010 \end{bmatrix}$$

- Add new word vector and apply ReLU:

$$h_3 = \max\left(0, \begin{bmatrix} 0.346 \\ -0.010 \end{bmatrix} + \begin{bmatrix} 0.20 \\ -0.10 \end{bmatrix}\right) = \max\left(0, \begin{bmatrix} 0.546 \\ -0.110 \end{bmatrix}\right) = [\mathbf{0.546}, \mathbf{0.0}]$$

Final Representation for Sentence 2: $[\mathbf{0.546}, \mathbf{0.0}]$

The Mathematical Proof

- **Sentence 1 Final Vector:** [0.258, 0.198]
- **Sentence 2 Final Vector:** [0.546, 0.0]

Even though both sentences contain the exact same words, **the final vectors are completely different!**

This mathematical walkthrough proves how the recurrent weight matrix (W_{hh}) and chronological sequence processing ensure that word order is permanently maintained.